MCS-26 - M1101 - LAB 02: PLAYFAIR CIPHER

Bangladesh University of Professionals (BUP)

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Exercise 2.1 - Matrix Construction

In a Playfair cipher, the 5x5 key matrix is built by taking a keyword, converting any 'J' to 'I', removing duplicate letters, and filling the remaining slots with the remaining letters of the alphabet (excluding 'J').

For the keyword 'CRYPTOGRAPHY':
1. Convert J to I (no J present): 'CRYPTOGRAPHY'
2. Remove duplicates: C, R, Y, P, T, O, G, A, H (Note: 'R', 'P', 'Y' are discarded as duplicates)
3. Fill rest of alphabet (excluding 'J'): B, D, E, F, I, K, L, M, N, Q, S, U, V, W, X, Z

This yields the following 5x5 key matrix, verified using the build_matrix() function:

C	R	Y	P	T
O	G	A	H	B
D	E	F	I	K
L	M	N	Q	S
U	V	W	X	Z

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Exercise 2.2 - Encryption Verification

We verify the step-by-step encryption of the plaintext 'HIDE THE GOLD IN THE TREE STUMP' using the keyword 'PLAYFAIR EXAMPLE'.

Key Matrix for 'PLAYFAIR EXAMPLE' (Duplicates removed, J replaced with I):

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

Detailed Digram Encryption Steps:

Digram	Matrix Positions	Playfair Rule	Encrypted
HI	H:(2,4), I:(1,0)	Rectangle (Swap Columns)	BM
DE	D:(2,2), E:(1,2)	Same Column (Shift Down)	OD
TH	T:(4,0), H:(2,4)	Rectangle (Swap Columns)	ZB
EG	E:(1,2), G:(2,3)	Rectangle (Swap Columns)	XD
OL	O:(3,2), L:(0,1)	Rectangle (Swap Columns)	NA
DI	D:(2,2), I:(1,0)	Rectangle (Swap Columns)	BE
NT	N:(3,1), T:(4,0)	Rectangle (Swap Columns)	KU
HE	H:(2,4), E:(1,2)	Rectangle (Swap Columns)	DM
TR	T:(4,0), R:(1,1)	Rectangle (Swap Columns)	UI
EX	E:(1,2), X:(1,3)	Same Row (Shift Right)	XM
ES	E:(1,2), S:(3,4)	Rectangle (Swap Columns)	MO
TU	T:(4,0), U:(4,1)	Same Row (Shift Right)	UV
MP	M:(1,4), P:(0,0)	Rectangle (Swap Columns)	IF

Final Ciphertext (spaces removed): BMODZBXDNABEKUDMUIXMMOUVIF

Exercise 2.3 - Digram and Statistical Analysis

a) Most Common English Digram in the Corpus

The most common English digram in our corpus is 'TH' with a relative frequency of 4.24%. This aligns with standard English language distributions where 'TH' is statistically the most dominant digram.

b) Digram Frequency in the Ciphertext

No, 'TH' is no longer the most common digram in the ciphertext. In fact, the most common ciphertext digram is now 'PD' with a frequency of only 2.92%. The highest peaks are drastically flattened, and the original statistical frequency characteristics are successfully spread out. This flat distribution demonstrates the effectiveness of the Playfair cipher's polyalphabetic substitution.

c) Index of Coincidence (IC) Evaluation

We calculated the Index of Coincidence (IC) for both the plaintext and ciphertext:

- Plaintext Index of Coincidence (IC): 0.06384
- Ciphertext Index of Coincidence (IC): 0.04781

Observation:

The plaintext IC (~0.0638) is extremely close to the expected value for standard English text (~0.0667), which reflects highly structured non-random letter distributions. The ciphertext IC (~0.0478) is significantly lower, moving towards the theoretical value of a purely random text (1/26 = 0.0385).

By replacing pairs of letters (digrams) instead of single letters, the Playfair cipher flattens the statistical frequency peaks, successfully defending against simple single-letter frequency attacks. The drop in IC serves as a mathematical proof of the increased randomness and security of the ciphertext compared to a monoalphabetic cipher like the Caesar cipher.

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PT-109 Decryption Walkthrough and Historical Context

During World War II, Lieutenant (and future US President) John F. Kennedy's boat, PT-109, was rammed and sunk by a Japanese destroyer in the Solomon Islands. A famous rescue mission ensued. A Playfair cipher message was transmitted by the Royal New Zealand Navy to coordinate search and rescue efforts.

We verified the decryption of this historical ciphertext using our implementation:
- Key: 'ROYAL NEW ZEALAND NAVY'
- Ciphertext:
  KXJEYUREBE ZWEHEWRYTU HEYFS KREHE GOYFI WTTTU OLKSY CAJPO BOTEI ZONTX BYBNT GONEY CUZWR GDSON SXBOU YWRHE BAAHY USEDQ

Decryption Matrix (ROYAL NEW ZEALAND NAVY):

R	O	Y	A	L
N	E	W	Z	D
V	B	C	F	G
H	I	K	M	P
Q	S	T	U	X

Decrypted Plaintext:

PTBOATONEOWENINELOSTINACTIONINBLACKESSSTRAITTWOMILESSWMERESUCOVEXCREWOFTWELVEXREQUESTANYINFORMATIONX

Analysis of Encryption / Decryption Errors:

Historically, the operator made spelling and encryption errors while encoding the message in the field. Our programmatic decryption perfectly reconstructs these exact anomalies:

1. 'OWENINE' instead of 'ONE O NINE':
   The word 'ONE' was incorrectly encrypted/decrypted, yielding 'OWENINE' which telegraphically reads as 'PT BOAT ONE O NINE'.

2. 'BLACKESSSTRAIT' instead of 'BLACKETT STRAIT':
   Because the double-T in 'BLACKETT' required duplicate padding during encryption, and due to operational slip-ups, the letters decrypted to 'BLACKESSSTRAIT' rather than 'BLACKETTSTRAIT'.

Despite these manual errors, the message was successfully decrypted by the rescue operators, and the crew of PT-109 was safely rescued. This illustrates both the real-world strength of the Playfair cipher in retaining message comprehensibility under field-encryption errors, and its susceptibility to manual operational mistakes.